Problem set 6

1) Double well potential

Consider a particle of mass m in a 1d potential $V(q) = \lambda(q^2-q_0^2)^2$.

Write down a path-integral in imaginary time that computes

$$\langle q_0, \tau/2 | -q_0, -\tau/2 \rangle = \langle q_0 | e^{-H\tau/\hbar} | -q_0 \rangle$$

- What's the Euclidean Lagrangian? How's it different from the real-time Lagrangian?
- Write down the Euler-Lagrange eq. for this Euclidean action.
- Compute the path-integral with this potential.

2) Free propagator

Derive the path-integral expression for the propagator a free particle in d spatial dimensions with the action $S_0 = \frac{1}{2} m^2 \int \dot{q}^2$ by discretizing time.

2) Path-integral for complex scalar

Consider the action for a complex massive free scalar field $\Phi(x)$.

- Compute $Z_0[0]$ explicitly.
- Write down the action in the presence of sources for both $\Phi(x)$ & $\Phi^*(x)$.

- Evaluate the path-integral & compute the 2-pt function by taking derivatives w.r.t. J & setting $J=0$.

- Show that differentiating the path integral gives time-ordered correlators.

3) Connected Correlators

Consider the generating functional $Z[J]$ for all correlators:

$$\frac{\delta}{i\delta J(x_1)} \cdots \frac{\delta}{i\delta J(x_n)} Z[J]\Big|_{J=0} = \langle \phi(x_1) \cdots \phi(x_n) \rangle$$

Show that if we define $W[J] = -i \log Z[J]$ differentiating $W[J]$ w.r.t. $i\delta J$ gives the connected correlators.